

Nypel® RC5035 BK

Polyamide 66

Nypel RC5035 BK is a 35% glass fiber reinforced, injection molding compound produced with 100% post-consumer recycled polymer content. This product is a regional grade available in North America only.

Applications

Nypel RC5035 BK is generally recommended for applications such as automotive components, industrial housings, caps, bolts, and racks

PHYSICAL	ISO Test Method	Property Value	
Density, g/cm ³	1183	1.45	
MECHANICAL	ISO Test Method	Dry	Conditioned
Tensile Modulus, MPa	527		
23°C		11,600	-
Tensile stress at break, MPa	527		
23°C		150	-
Tensile strain at break, %	527		
23°C		1.7	-
Flexural Modulus, MPa	178		
23°C		10,500	-
IMPACT	ISO Test Method	Dry	Conditioned
Izod Notched Impact, kJ/m²	180		
23°C		7.4	-
Charpy Notched, kJ/m²	179		
23°C		7.4	-
THERMAL	ISO Test Method	Dry	Conditioned

Melting Point, °C	3146	256	-
HDT A, ° C	75	236	-

Processing Guidelines

Material Handling

Max. Water content: 0.15%

Material is supplied in sealed containers and drying prior to molding in a dehumidifying or desiccant dryer is recommended. Drying parameters are dependent upon the actual percentage of moisture in the pellets and typical pre-drying conditions are 2-4 hours at 180F (83C). Further information concerning safe handling procedures can be obtained from the Safety Data Sheet (MSDS), or by contacting your BASF representative.

Typical Profile

Rear: 265-277°C (509-530°F)

Middle: 277-288°C (530-550°F)

Front: 288-300°C (550-572°F)

Nozzle: 290-304°C (554-579°F)

Melt: 280-305°C (536-581°F)

Mold Temperature: 80-90°C (176-194°F)

Injection and Packing Pressure: 35-125 bar (500-1500 psi)

Mold Temperatures

A mold temperature of 80-90°C (176-194°F) is recommended, however temperatures of as low as 45°C (113°F) and as high as 105°C (221°F) can be used where applicable.

Pressures

Injection pressure controls the filling of the part and should be applied for 90% of ram travel. Packing pressure affects the final part and can be used effectively in controlling sink marks and shrinkage. It should be applied and maintained until the gate area is completely frozen off.

Back pressure can be utilized to provide uniform melt consistency and reduce trapped air and gas. Minimal back pressure should be utilized to prevent glass breakage.

Fill Rate

Fast fill rates are recommended to ensure uniform melt delivery to the cavity and prevent premature freezing. Surface appearance is directly affected by injection rate.

Note

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